University of Milano-Bicocca

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Deep Learning in Healthcare: Dementia diagnosis through GNN

Supervisor: Paolo Napoletano

Co-supervisor : Ionel Eduard Stan

Master’s degree thesis by:

Alfio Leanza

student ID number 910235

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Abstract

Neurodegenerative dementias, particularly Alzheimer's disease and frontotemporal dementia, pose a clinical challenge even for differential diagnosis in the early stages. In this context, the electroencephalogram (EEG) is a non-invasive and relatively inexpensive tool capable of reflecting functional alterations through spectral and network changes. This thesis proposes and evaluates a deep learning pipeline for multi-class classification (healthy controls, **HC**; frontotemporal dementia, **FTD**; Alzheimer's disease, **AD**) from resting-state EEG, combining a time-frequency representation based on Continuous Wavelet Transform (CWT) and a Graph Neural Network (GNN) model that interprets EEG channels as nodes in a graph.

The experiments are conducted on the public OpenNeuro dataset ds004504 using a subject-independent split to avoid leakage between training/validation/testing. The main contribution consists in reformulating the classification head using the One-vs-Rest (OVR) scheme, as an alternative to multiclass Softmax, while keeping the GNN backbone unchanged. The results indicate that OVR improves generalization stability on the test set and that the benefit is amplified at the subject level thanks to an aggregation of predictions computed on short, non-overlapping EEG segments (crop) using weighted soft voting. Finally, an exploratory experiment to estimate reliability is discussed, outlining limitations and possible future developments.

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1 Introduction

Neurodegenerative dementias are one of the leading causes of disability in the aged population and have an increasing impact on healthcare systems and society. Among the most relevant forms are Alzheimer's disease (AD), the most common, accounting for approximately 60-80% of dementia cases [1] and frontotemporal dementia (FTD), less common (approximately 5-10%) but relatively more prevalent in younger age groups and often with a presenile onset [2]. Although AD and FTD share a progressive nature and involve widespread cortical networks, they differ in clinical pattern, onset location, and neuropathological mechanisms; an accurate differential diagnosis is therefore crucial for prognosis, therapeutic management, and access to targeted trials.

The diagnostic process, proposed by the NIA-AA guidelines, integrates medical history, neuropsychological assessments, and, when available, imaging and cerebrospinal fluid biomarkers [3]. However, especially in the early stages, distinguishing between AD and FTD can be complex: symptoms can be subtle, overlap with other conditions, or vary between individuals, resulting in a risk of misclassification. In this scenario, the electroencephalogram (EEG) is a potentially valuable tool: it is non-invasive, relatively inexpensive, and capable of describing the dynamic functioning of cortical networks. The literature reports, for example, a slowing of the trace in AD (increase in delta/theta power and reduction in alpha/beta) and more heterogeneous patterns in FTD, suggesting that spectral and relational measures may contain discriminatory information useful for automatic diagnosis [4].

In the last few years, the availability of public EEG datasets and the maturity of machine learning and deep learning techniques have encouraged the development of data-driven models for diagnostic classification. In this context, a line particularly suited to EEG has emerged: Graph Neural Networks (GNN), which model channels as nodes in a graph and capture information not only "per channel" but also in the relationships between sensors. This choice is consistent with a "network" interpretation of neurodegenerative diseases and with the inherently multichannel nature of EEG.

Within this context, this thesis focuses on the public benchmark ds004504, available on OpenNeuro, which includes healthy subjects (HC), AD patients, and FTD patients in a standardized format (EEG-BIDS) and with a number of channels typical of clinical EEG (19 electrodes). The objective is to evaluate whether a graph-centric pipeline, fed by a rich time-frequency representation (Continuous Wavelet Transform, CWT), is able to discriminate between the three classes in a subject-independent setting, i.e., avoiding that crops of the same subject appear in different splits. This constraint is essential in EEG: segmentation produces many samples per subject, and without subject-level splits, performance can be artificially inflated by "identity" leakage.

From a methodological point of view, the proposed pipeline transforms each EEG into a multi-channel CWT scalogram (40 frequencies in the range 0.5-45 Hz), then constructs a sensor graph with sparse and neuro-plausible topology; a GNN backbone produces a global representation of the crop via local encoder, message passing, and graph-level readout (pooling). The backbone architecture is taken as the baseline provided in the working context; the central contribution of the thesis concerns the reformulation of the classification head: instead of the standard multiclass formulation with Softmax, a One-vs-Rest (OVR) head is proposed, which breaks down the problem into binary classifiers 'class vs rest'. This choice aims to make class optimization more manageable (an important aspect in the presence of imbalance and phenomenological overlap, particularly for FTD) and to improve robustness in generalization.

Since the clinical decision unit is the subject and not the individual signal second, the thesis also systematically evaluates subject-level aggregation: predictions performed on individual crops (crop-level) are integrated through weighted soft voting, in which the most confident crops contribute most to the final decision. Experimental results show that, keeping the backbone fixed, the combination of OVR + weighted aggregation amplifies the diagnostic evidence distributed over time and produces a significant advantage over the Softmax baseline at the subject level. In addition, a complementary idea is explored: a monitor that attempts to estimate the reliability of the main model's prediction and use it as a signal to further tune the aggregation weights; the experiment is discussed as an exploratory direction, highlighting its potential and critical issues.

2 Related Works

A growing number of studies have leveraged public EEG datasets and modern computational resources to develop machine learning and deep learning methods for the automated diagnosis of dementia. Traditional approaches extract qEEG features (e.g., band power, coherence) and use them as input for classifiers such as Support Vector Machine (SVM), Random Forest, or boosting methods.

More recent models exploit Convolutional (CNN), Recurrent (RNN), and, above all, Graph Neural Networks (GNN) architectures, which represent the EEG as a graph in which the nodes correspond to the electrodes and the edges encode anatomical or functional relationships. This modeling is particularly well suited to EEG, which originates as a spatiotemporal measurement from a network of sensors on a 2D surface (the scalp) but reflects three-dimensional cortical interactions.

GNNs enable joint learning:

- spectral patterns (e.g., slowdown in slow bands);
- spatial patterns (topographic distribution of changes);
- relational patterns (connectivity between frontal, temporal, parietal regions, etc.).

In this context, the objective of diagnosing dementia using GNN fits into the field of Deep Learning in Healthcare, integrating structured neurophysiological data (in standard formats such as EEG-BIDS) with graphical models capable of exploiting sensor topology and time-frequency representations of the signal.

2.1 GNN for EEG: graph construction strategies

In EEG-based studies, the graph can be defined in different ways, each with methodological and interpretative implications:

- **Topological/anatomical graph (sensor graph):**

The nodes are electrodes and the arcs are defined by spatial proximity or predefined connections (e.g., fronto-central chains, interhemispheric homologues). This approach imposes a stable inductive bias and reduces the risk of overfitting to very dense graphs, but does not directly incorporate subject-specific functional connectivity;

- **Functional connectivity graph:**

  The arcs are weighted using measures such as coherence, PLV/PLI, mutual information, etc. In this case, GNN operates on an explicit estimate of the functional network. However, performance can depend significantly on the connectivity measure, preprocessing, and time window chosen. The idea that the choice of connectivity is a “bottleneck” has been highlighted in comparative studies on EEG graphs for AD;

- **Multi-band/multi-view graphs:**

  Connectivity is estimated separately for each band or for different coupling definitions, resulting in multiple graphs per subject (or per window). Multi-graph or multi-layer approaches attempt to integrate these views, at the cost of greater modeling complexity.

These strategies reflect a common trade-off: more “informative” graphs (weighted connectivity) can capture more direct network biomarkers but introduce an intermediate estimate that can amplify noise and dependencies on hyperparameters; fixed topological graphs are more stable and controllable but require the model to implicitly learn functional relationships.

2.2 OpenNeuro ds004504 dataset as a benchmark for AD-FTD-HC

A key contribution to the community is the ds004504 dataset, which includes 19-channel closed-eye resting-state recordings with three classes (HC, FTD, AD) and is available in

EEG-BIDS format and on OpenNeuro (ds004504) [5] [6]. The public availability of a dataset with FTD (in addition to AD and controls) is particularly relevant because many public EEG resources are biased towards AD vs HC or MCI vs HC, while the AD-FTD distinction is clinically crucial.

The original work on this dataset proposes a benchmark based primarily on qEEG features (relative band power) and classical models, providing a reproducible starting point on which subsequent research has built extensions in the direction of deep learning and network models.

Work on ds004504: from feature engineering to network methods

Subsequent works using the dataset can be grouped into three families:

- **Feature-based and region/biomarker-based methods:**
  Some studies explore features that are richer than just band power (complexity, entropies, nonlinear indices) and selection/interpretation strategies to highlight more informative regions. This line of research also includes studies that evaluate the informativeness of specific lobes or subnetworks, with the aim of linking classification to interpretable neurophysiological biomarkers;
- **Non-graphical deep learning on ds004504:**
  Convolutional models optimized for EEG and modern variants (e.g., ConvNeXt-like architectures adapted to the EEG domain) have been proposed for automatic AD/FTD classification on ds004504. These works show that high performance can be achieved, but comparability strongly depends on the protocol (segmentation, subject-independent split, subject-level aggregation) and the risk of leakage if windows from the same subject end up in different sets;
- **Graphical and connectivity approaches on ds004504:**
  In previous years, approaches have emerged that construct graphs (topological or connectivity) and apply network models or GNNs. A recent example is a dual-path GNN framework for dementia diagnosis, which explicitly cites the Miltiadous

dataset as an experimental reference and proposes a graphical pipeline to capture complementary patterns. Another direction is the use of effective/functional connectivity graphs and lightweight or causal-oriented GNN architectures to distinguish AD and FTD, aiming to integrate inter-channel relationships more directly. There are also proposals that combine graph domain transformations with classifiers ("graph spectral" approaches) applied that data, with an emphasis on regional interpretation and connection weighting.

Overall, the literature on the dataset suggests that graphical models become particularly useful when:

- discrimination between classes depends not only on local power, but also on the relational pattern between areas (frontotemporal vs. posterior, asymmetries, decouplings);
- greater interpretability in terms of nodes/arcs and a natural link to network biomarkers is desired.

2.3 Relation to prior work

Given the previous work, this thesis suggests a pipeline that highlights three aspects:

- **High-resolution time-frequency representation:**
  Instead of compressing the information a priori into a few bands (RBP) or manually selected features, a wavelet representation (CWT scalograms) is used to preserve peaks and non-stationary patterns in the time-frequency domain;
- **Graph-centric modeling with neuro-plausible topology:**
  The electrode graph is defined in a sparse and plausible way, reducing density and potential undesirable effects on complete graphs; relational information is learned through message passing;
- **Subject-independent protocol and subject-level decision:**

Since the clinical unit is the subject, the use of subject-independent splits and downstream aggregation (voting) allows for a more realistic assessment of generalization compared to sample-independent settings.

To sum up, the proposed approach aims to combine the advantages of qEEG literature (focus on known pathological patterns) with the advantages of graphical models (network modeling) and time-frequency representations (richness of information), while remaining consistent with the ds004504 benchmark and its EEG-BIDS structure.

3 Preliminary concepts

This chapter provides the minimum theoretical basis necessary to understand the proposed methodology: the nature of the EEG signal and its representations in the frequency and time-frequency domains; essential elements of graph theory; principles of Graph Neural Networks (GNN) as message passing models. The implementation details (specific architecture, OVR, voting, training) will be discussed in the Methodology chapter, with only the fundamental concepts being covered here.

The following notation is used below:

- f_s: sampling frequency (Hz);
- C: number of channels/electrodes;
- d: crop duration (in seconds);
- $N = d \cdot f_s$: number of samples per crop;
- F: number of scales/frequencies in the CWT representation;
- $\mathbf{X}^{(k)} \in \mathbb{R}^{C \times N}$: $k - th$ crop in the time domain;
- $\mathbf{W}_c^{(k)} \in \mathbb{C}^{F \times N}$: CWT coefficients of channel c;
- $A_c^{(k)} =| \mathbf{W}_c^{(k)} |\in \mathbb{R}^{F \times N}$: channel c amplitude scale;
- $\mathbf{A}^{(k)} \in \mathbb{R}^{C \times F \times N}$: multi-channel crop scale diagram;
- $\mathcal{D} = \{(\mathbf{X}^{(k)}, y^{(k)})\}_{k=1}^{M}$: supervised dataset of Mcrops, with label $y^{(k)}$.

In order to give context, each recording is sampled at $f_s = 500\,Hz$ and acquired with $C = 19$ electrodes; segmenting into non-overlapping crops of $d = 1s$ it returns in $N = 500$, represented as:

$$\mathbf{X}^{(k)} \in \mathbb{R}^{19 \times 500}$$

where k indexes the crop instances extracted from all subjects.

Defining $F = 40$, for crop k and channel c, the CWT returns:

$$\mathbf{W}_c^{(k)} \in \mathbb{C}^{40\times500}$$

Finally, using as input feature $\boldsymbol{A}_c^{(k)} = |\mathbf{W}_c^{(k)}| \in \mathbb{R}^{40\times500}$ and stacking the C channels give

$$\boldsymbol{A}^{(k)} \in \mathbb{R}^{19\times40\times500}$$

After this, the supervised dataset:

$$\mathcal{D} = \{(\mathbf{X}^{(k)}, y^{(k)})\}_{k=1}^{M} \quad (equivalently\ \mathcal{D} = \{(A^{(k)}, y^{(k)})\}_{k=1}^{M}\ after\ CWT)$$

where M is the total number of crops, depending on the duration of each subject recording, and $y^{(k)} \in \{HC, FTD, AD\}$ is the label inherited from the subject.

3.1 EEG, PSD and band power

The EEG measures cortical electrical activity resulting from the sum of postsynaptic potentials generated by populations of pyramidal neurons, recorded via electrodes placed on the scalp according to standard conventions (e.g., the international 10-20 system). The signal is often described in terms of frequency bands:

- **Delta**: 0.5-4 Hz;
- **Theta**: 4-8 Hz;
- **Alpha**: 8-13 Hz;
- **Beta**: 13-25 Hz;
- **Gamma**: 25-45 Hz.

These bands reflect functional configurations of large neural networks and vary with age, alertness, and pathological conditions. In resting-state conditions with eyes closed, in most healthy adults the dominant rhythm is typically posterior alpha, an expression of synchronized occipito-parietal activity, accompanied by theta and beta components of lesser amplitude.

An “average” frequency characterization of the signal is obtained by estimating the power spectral density (PSD). In practice, given a discrete signal $x[n]$, the PSD can be estimated using a periodogram or more robust methods such as Welch, which average estimates over overlapping segments [7].

The **power** of a band, denoted by the frequency band $[f_1, f_2]$, is obtained by integrating (or summing in discrete form) the PSD over that interval. To compare subjects with different scales, relative band power is often used, normalizing the band power for the total power in a reference interval (e.g., 0.5-45 Hz). This representation is compact and interpretable, but it does not explicitly describe time-frequency variations and depends on the assumption of (almost) stationarity on the analyzed window.

3.2 Continuous Wavelet Transform and Scalograms

For non-stationary signals such as EEG, a joint time-frequency representation is often more informative. The Continuous Wavelet Transform (CWT) of a continuous signal $x(t)$ with respect to a mother wavelet $\psi(t)$ is defined as [8]:

$$W_x(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} x(t)\ \psi^*\left(\frac{t-b}{a}\right) dt$$

where:

- $a > 0$ is the scale (inverse to the frequency);
- b is the temporal translation;
- $\psi^*\left(\frac{t-b}{a}\right)$ indicate the conjugate complex.

Complex Morlet wavelets are frequently used in EEG, offering an effective compromise between temporal and frequency resolution for near-sinusoidal oscillatory components (alpha/beta). The scalogram is obtained as the modulus of the coefficients:

$$A_x(a,b) = | W_x(a,b) |$$

Normalization

To stabilize learning and reduce inter-subject scale differences, normalization is often applied. A simple and leakage-free choice is the z-score on the single crop:

$$\tilde{A}^{(k)} = \frac{A^{(k)} - \mu^{(k)}}{\sigma^{(k)} + \varepsilon}$$

where $\mu^{(k)}$ and $\sigma^{(k)}$ are statistics calculated on the crop coefficients (globally or per channel) and $\varepsilon > 0$ avoids numerical instability.

3.3 Graph theory: essential definitions

A graph is a pair $G = (V, E)$, where:

- V is the set of nodes;
- $E \subseteq V \times V$ is the set of arcs (oriented or unoriented).

In the EEG case, a natural representation is a sensor graph:

- Each electrode corresponds to a node $v_i \in V$;
- Arcs represent relationships between electrodes (topographical proximity, interhemispheric homology, predefined connections).

The adjacency matrix is defined as $\mathbf{A} \in \mathbb{R}^{|V| \times |V|}$, where $A_{ij} \neq 0$ indicate a connection between i and j. In modern implementations, it is common to describe E using a list of edges (edge_index) compatible with graph learning libraries [9].

3.4 Graph Neural Networks: message passing and global readout

Graph Neural Networks (GNN) generalize neural networks to structured data such as graphs, updating node representations by aggregating information from neighbors (message passing) [10]. A general form of update to layer l is:

$$\mathbf{h}_i^{(l+1)} = \phi\left(\mathbf{h}_i^{(l)}, \mathrm{AGG}_{\, j \in \mathcal{N}(i)}\, \psi\left(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, e_{ij}\right)\right)$$

where:

- $\mathbf{h}_i^{(l)}$ is the embedding of node i;
- $\mathcal{N}(i)$ is the neighborhood of i;
- e_{ij} are any attributes/weights of the arc;
- ψ is the message function;
- AGG is a permutation-invariant aggregator (sum/average/max);
- ϕ is a transformation (e.g., linear + nonlinearity).

A common practical formulation (graph with arc weight α_{ji}) is:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}\ \mathbf{W}^{(l)} \mathbf{h}_j^{(l)}\right)$$

where $\mathbf{W}^{(l)}$is a weight matrix and $\sigma(\cdot)$ an activation.

3.4.1 Readout for graph classification

In problems where the label is associated with the entire graph (here: a crop/graph), after L layers the node embeddings are aggregated into a global vector $\mathbf{g}$ via pooling:

$$\mathbf{g} = \mathrm{POOL}_{\,i \in V}\, \mathbf{h}_i^{(L)}$$

with $\mathrm{POOL} \in \{\text{sum,mean,max}\}$. Vector $\mathbf{g}$ then feeds a standard classifier (Multi-Layer Perceptron, MLP).

3.4.2 Known limitations: oversmoothing and bottleneck

The depth of message passing and the density of the graph can introduce undesirable phenomena:

- **oversmoothing**, so that embeddings of different nodes tend to align;
- **bottleneck/oversquashing**, in which a lot of information is compressed along a few paths, degrading the ability to propagate useful signals on complex graphs.

These aspects are discussed in the literature and often motivate the use of graphs that are not excessively dense and a limited number of layers [11] [12].

3.4.3 Reminders on stabilization and optimization

The following are common in network training:

- **Batch Normalization**, which normalizes internal activations and facilitates convergence [13];
- **Adam**, a widely used adaptive optimizer thanks to parameter updating based on gradient moment estimates [14].

These tools will be referred into the Methodology chapter to justify implementation choices.

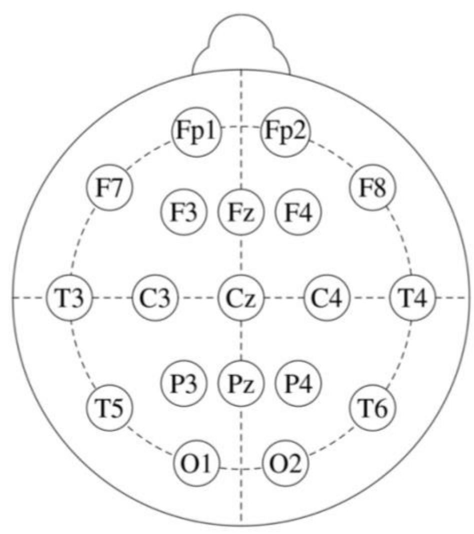
4 Dataset description and processing from EEG to CWT representations

The experimental work in this thesis is based on the ds004504 dataset, which contains routine resting-state EEG recordings with eyes closed acquired in a clinical setting at the **2nd Department of Neurology of AHEPA University Hospital** (Thessaloniki, Greece). The cohort includes:

- **36** patients with **Alzheimer's disease (AD)**;
- **23** patients with **frontotemporal dementia (FTD)**;
- **29 healthy control** subjects **(HC)**, matched for age.

with demographic information and MMSE score per subject.

Recordings are made using a Nihon Kohden clinical system and include **19 electrodes** on the scalp according to the 10-20 system:



Reproduced from [15].

There are also mastoid electrodes (A1, A2) used in the reference/re-referencing pipeline in the derivative. The sampling frequency is **500 Hz**, and the recordings have variable duration (typically minutes), sufficient to characterize the resting state.

4.1 EEG-BIDS structure and pre-processed derived data

The dataset is organized according to the EEG-BIDS standard, an extension of the Brain Imaging Data Structure for electroencephalographic data [16]. Specifically, the structure includes:

- a dataset_description.json file with general metadata;
- a participants.tsv table with subject IDs, diagnoses, ages, MMSE scores, etc;
- sub-0XX/ folders with acquisition files and EEG signals (EEGLAB *.set format) [17];
- a derivatives/ folder with pre-processed versions of the recordings, organized using the same logic by subject.

The adoption of EEG-BIDS makes the workflow reproducible and facilitates the automated loading of signals and metadata via analysis libraries (e.g., MNE) [18].

4.2 Pre-processing pipeline in derivatives

It's suggested using derivatives data, which is obtained through a standardized process. The main steps are:

1. **Band-pass filtering (0.5-45 Hz):**
   Band-pass filter (Butterworth) consistent with classic EEG bands and with qEEG use in dementia, removing very low frequency drift and non-informative high frequency noise.
2. **Re-referencing:**
   After filtering, the signal is re-referenced to the **A1-A2** average, obtaining a bilateral reference and reducing dependence on a single electrode.
3. **Artifact Subspace Reconstruction (ASR):**
   ASR is an automatic procedure that identifies segments with abnormal standard deviation compared to a "clean" model and mitigates their impact through correction in the component space.
   In the dataset: 0.5 s windows and threshold 17 (conservative value).

4. **ICA+ICLabel:**

   Independent Component Analysis (ICA), RunICA, is performed and the independent components are automatically classified using ICLabel; components attributed to ocular/muscular artifacts are removed [19]. This step is also relevant with closed eyes, because micro saccades, partial blinks, and muscle tension may persist.

5. **Baseline correction:**

   Baseline correction to bring the signal average back to zero and reduce residual drift.

The result is a set of denoised and normalized tracks, suitable for segmentation and data-driven analysis.

4.3 Thesis input data: loading, conversion and segmentation into crops

In this thesis pre-processed recordings (derivatives) are used, but feature segmentation and representation are redefined for training a GNN.

The pipeline (based on tools compatible with EEG-BIDS and EEGLAB formats):

- associate each subject sub-0XX with the clinical information in participants.tsv;
- load the pre-processed continuous EEG file *.set;
- select and sort the **19 channels** of interest (Fp1-O2), discarding any auxiliary channels;
- converts amplitudes into **microvolts (µV)**, in line with clinical conventions;
- save data in float32 (trade-off between precision and memory).

Segmentation

Given a continuous trace lasting T seconds, non-overlapping crops with a duration of $d = 1$s. With a sampling frequency of $f_s = 500$Hz, each crop contains

$$N = d \cdot f_s = 500$$

samples per channel. The number of crops per subject is therefore:

$$M_i = \left\lfloor \frac{T_i}{d} \right\rfloor$$

Each crop is represented as:

$$\mathbf{X}^{(k)} \in \mathbb{R}^{C \times N}, C = 19,\ N = 500, k = 1, \dots, M_i$$

Each of them inherits the subject's diagnostic label (HC/FTD/AD), allowing to work at the crop-level and then aggregate predictions at the subject-level.

4.4 From the time series to the CWT representation

To obtain a rich representation in the time-frequency domain (as described in the chapter “Preliminary Concepts”), the **Continuous Wavelet Transform (CWT)** is calculated for each crop $\mathbf{X}^{(k)}$.

Additionally, the following are selected:

- $F = 40$ scales/frequencies covering the range **0.5-45 Hz** (approximately log-spaced);
- CWT evaluation on $N = 500$ time points of the crop (without subsampling on the CWT time).

Consequently, the final form for crop is:

$$A^{(k)} \in \mathbb{R}^{19 \times 40 \times 500}$$

Depending on the implementation conventions, the tensor can be reordered (e.g., 40,500,19) without changing the information it contains.

4.5 Comparison with the original benchmark

In the original benchmark, a pipeline based on **Relative Band Power (RBP)** features is defined, with:

- 4 s windows with 50% overlap;
- PSD estimate (Welch);
- calculation of average power in delta-gamma bands;
- normalization in RBP (bandwidth/total ratio 0.5-45 Hz);
- classification with classical models in leave-one-subject-out validation.

In this thesis, the pipeline is reinterpreted in terms of deep learning:

- a compact feature vector (RBP) is replaced with a **high-dimensional CWT** representation per channel;
- segmentation into **1-second crops** without overlap is adopted to increase the number of examples and temporal granularity;
- The EEG is modeled as an **electrode graph**, and a GNN is tasked with learning discriminative space-spectral combinations.

These choices aim to reduce the imposition of a priori assumptions on bands and regions (e.g., “posterior alpha” as the only dominant information), leaving the model free to learn relevant patterns by simultaneously exploiting sensor topology and time-frequency dynamics.

5 Methodology

This chapter describes in an operational and formal way the experimental pipeline adopted for diagnostic classification from EEG (HC/FTD/AD) in a subject-independent setting. The basic concepts (CWT, scalograms, message passing in GNNs, and notation) have already been introduced in previous chapters; here, instead, we emphasize the **implementation choices**, the **model structure**, and the **technical motivation** behind the proposed extensions.

The approach follows a “graph-centric” approach: each EEG crop is transformed into a multi-channel time-frequency representation and then encoded as a **sensor graph**; a GNN produces a global representation of the crop, and a classification head generates the prediction. The backbone (baseline) architecture was provided as a starting point within the group's line of work; the main contribution of this thesis is to have explored architectural and training variants until converging on a **One-vs-Rest (OVR)** reformulation of the classification head, which is formalized and motivated here.

5.1 Data construction: from EEG crop to sensor graph

Segmentation defines the learning unit. Each recording is divided into crops of duration d seconds; with sampling frequency f_s, each crop contains:

$$N = d \cdot f_s$$

samples per channel. Let C the number of channels (electrodes), then the $k-th$ crop in the time domain is represented as:

$$\mathbf{X}^{(k)} \in \mathbb{R}^{C \times N}$$

The CWT is calculated on each channel and the amplitude scalogram is used as a feature; for channel c of crop k, we obtain:

$$A_c^{(k)} =| \mathbf{W}_c^{(k)} |\in \mathbb{R}^{F\times N}$$

where F is the number of scales/frequencies sampled in the interval of interest. Stacking the channels:

$$\boldsymbol{A}^{(k)} \in \mathbb{R}^{C\times F\times N}$$

Since the context is subject-independent, normalization is set to avoid contamination between splits: per-crop standardization (or per-channel within the crop, depending on the variant) is applied directly to the time-frequency representation, in the form:

$$\tilde{\mathrm{A}}^{(k)} = \frac{\boldsymbol{A}^{(k)} - \mu^{(k)}}{\sigma^{(k)} + \varepsilon}$$

The representation $\tilde{\mathrm{A}}^{(k)}$ maintains high dimensionality; to make it manageable without destroying the time-frequency structure, deterministic block temporal compression is introduced. The time axis is partitioned into B consecutive equal blocks, and the average is calculated within each block:

$$\bar{A}_{k,i}(:,b) = \frac{1}{N/B} \sum_{t=(b-1)\frac{N}{B}+1}^{b\frac{N}{B}} \tilde{A}_{k,i}\,(:,t), \qquad b = 1,\dots,B$$

obtaining for each electors i a matrix $\bar{A}_{k,i} \in \mathbb{R}^{F\times B}$. The node feature is then obtained through vectorization:

$$\mathbf{x}_{k,i} = \mathrm{vec}(\bar{A}_{k,i}) \in \mathbb{R}^{FB}$$

In this way, spectral information is preserved along F, while temporal dynamics are conserved at low resolution along B; furthermore, the FB dimensionality is compatible with a lightweight neural encoder and batch training.

Each crop is finally represented as a graph $G = (V, E)$, with $|V| = C$ nodes (one per sensor) and edges E defined by a sparse topology shared among the samples. The choice of a sparse topology introduces an inductive bias consistent with the sensor layout and reduces critical issues typical of message passing on very dense graphs (degradation of the information signal and instability). In operational terms, the entire dataset is then viewed as a collection of graphs with nodal features $\mathbf{x}_{k,i}$ and constant structure E.

5.2 Backbone provided: local encoder, weighted message passing and global readout

The starting architecture is a GNN backbone provided as a baseline: the thesis does not intervene on this structure in its fundamental components but assumes it as the "engine" for extracting representations. The logic of the backbone is to clearly separate the learning of local patterns (per electrode) from the relational combination on the graph.

The first stage is an encoder shared between nodes, implemented as an MLP with non-linearity, normalization, and dropout, which projects the feature $\mathbf{x}_{k,i}$ into a latent space:

$$\mathbf{h}_{k,i}^{(0)} = \phi(\mathbf{x}_{k,i}), \mathbf{h}_{k,i}^{(0)} \in \mathbb{R}^{d_0}$$

This transformation learns nonlinear combinations of intra-channel time-frequency patterns prior to mixing between sensors.

This is followed by a message passing block based on the GraphConv layer. The distinguishing feature of the baseline is the use of **learnable edge weights** α_{ij}, one per edge (directed), which modulate the intensity of the neighbors' contribution.

A layer can be expressed in the form:

$$\mathbf{h}_{k,i}^{(l+1)} = \sigma\left(\mathbf{W}_{\text{self}}^{(l)}\mathbf{h}_{k,i}^{(l)} + \sum\nolimits_{j\in\mathcal{N}(i)} \alpha_{ij}\ \mathbf{W}_{\text{neigh}}^{(l)}\mathbf{h}_{k,j}^{(l)}\right)$$

where $\mathcal{N}(i)$ is the neighborhood of i and $\sigma(\cdot)$ is a ReLU. This provides a controlled compromise between a completely “static” graph and more complex content-dependent attention schemes: the structure is fixed, but the relative importance of the connections is optimized by the data with a number of parameters proportional to $\mid E \mid$, which is generally stable even in scenarios with limited numbers.

After layer L, embeddings for node $\mathbf{h}_{k,i}^{(L)}$ are obtained. Since the prediction is associated with the crop (graph) and not with the individual sensor, a permutation-invariant readout (global pooling) is applied to obtain a unique representation of the sample:

$$\mathbf{g}_k = \mathrm{POOL}(\{\mathbf{h}_{k,i}^{(L)}\}_{i=1}^{C})$$

The vector $\mathbf{g}_k$ constitutes the output of the backbone and feeds the classification head. It is on this final part—leaving the graph-centric extractor unchanged—that the main contribution of the thesis is inserted.

5.3 One-vs-Rest (OVR) head as a reformulation of the problem

Experimental work has shown that, in a subject-independent setting with potentially unbalanced classes, the standard multiclass formulation can be sensitive to the competition imposed by softmax. For this reason, after numerous experiments involving regularization choices and head variants, the consolidated contribution was the adoption of a One-vs-Rest (OVR) head, which reformulates the multiclass classification as a set of K binary tasks, while keeping the backbone identical.

In the standard multiclass case, the head produces K logit $\mathbf{z}_k \in \mathbb{R}^K$ and the distribution is obtained via softmax:

$$p(y = r \mid \mathbf{g}_k) = \frac{\exp\,(z_{k,r})}{\sum_{u=1}^{K} \exp\,(z_{k,u})}$$

The associated loss is cross-entropy:

$$\mathcal{L}_{\text{softmax}} = -\frac{1}{\mid \mathcal{B} \mid} \sum_{k \in \mathcal{B}} \log\, p(y = y^{(k)} \mid \mathbf{g}_k)$$

with $\mathcal{B}$ training batches. This scheme forces direct competition between classes: increasing one probability implies decreasing the others, and the gradient is distributed in a "paired" manner between classes.

In the OVR formulation, on the other hand, starting from the representation of the sample $\mathbf{g}_k$ (o or from its projection $\mathbf{f}_k$), K independent binary classifiers are defined. For each class r, a binary target is introduced:

$$y_r^{(k)} = \begin{cases} 1 & \text{if } \mathrm{y}^{(k)} = r \\ 0 & \text{otherwise} \end{cases} \qquad r = 1, \dots, K,$$

a scalar K logits are calculated:

$$z_{k,r} = \mathbf{w}_r^\top \mathbf{f}_k + b_r, s_{k,r} = \sigma(z_{k,r}) = \frac{1}{1 + e^{-z_{k,r}}}.$$

The loss becomes the sum (or average) of the binary cross-entropy:

$$\mathcal{L}_{\text{OVR}} = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \sum_{r=1}^{K} (-\mathrm{y}_r^{(k)} \log\, s_{k,r} - (1 - y_r^{(k)}) \log\,(1 - s_{k,r})).$$

Since OVR outputs are not normalized, multiclass prediction is reconstructed by imposing a unique choice using a winner-takes-all rule:

$$\hat{y}^{(k)} = \arg \max_{r \in \{1,\dots,K\}} s_{k,r}.$$

There are two technical reasons why OVR can be advantageous in this context. On the one hand, it decouples margin optimization: each head learns "class r vs. rest" separately, reducing the effect of global competition when one class has a more nuanced or intermediate profile than the others. On the other hand, the binary formulation makes it natural to introduce balancing strategies (weights per class/positives or specific thresholds) without altering the backbone structure, allowing for more direct control of the trade-off between false positives and false negatives for each class.

Training is conducted in a manner consistent with the subject-independent objective: the train/validation/test split is defined at the subject level so that all crops from the same subject belong to the same set, avoiding leakage. The training optimizes $\mathcal{L}_{\text{softmax}}$ in the baseline and $\mathcal{L}_{\text{OVR}}$ in the proposed model; in both cases, regularization via dropout and weight decay is used to contain overfitting, and checkpoint selection is performed on validation using the main metric defined in the Results chapter. From a computational point of view, the PyTorch Geometric library allows efficient batching of homogeneous graphs [9]: the nodal features are concatenated along the node dimension, and the pooling operator reconstructs, for each graph in the batch, the corresponding global representation $\mathbf{g}_k$ on which Softmax or OVR operate.

In summary, the methodology maintains a robust and well-established graph-centric backbone as a baseline, focusing its contribution on a reformulation of the classification head and objective function. This choice aims to improve stability and generalization capacity in the subject-independent setting, without introducing unnecessary structural complexity in the feature extraction part.

5.4 Exploratory experiment: Monitor

One of the first directions explored during the thesis, prior to the formalization of the One-vs-Rest head, stems from an observation that is seemingly simple but methodologically relevant: not all predictions at the crop level have the same reliability, and this reliability does not necessarily coincide with the numerical "certainty" returned by the maximum probability. In particular, the analysis of probability distributions, which will be described in the next chapter, shows that a significant proportion of very confident predictions may be incorrect (for example, for OVR, a fraction of crops with $p_{\text{top}} \geq 0.9$ remain incorrect), suggesting that the native confidence of the classifier, while informative, is not an exhaustive descriptor of error [20] [21].

Hence the idea: to introduce a second model, called a monitor, with the sole purpose of predicting whether the main model's decision on that crop is correct or incorrect. In other words, instead of asking the monitor "what clinical class is it?", we ask "is this prediction reliable?”. The ambition is not to replace the classifier, but to estimate a second-level confidence, closer to a probability of correctness, by subsequently weighting or discarding the crops for aggregation by subject.

5.4.1 Definition of the Good/Bad targets and construction of the Monitor dataset

Formally, given a crop k with true label $y^{(k)}$ and main model prediction $\hat{y}^{(k)}$, a binary variable is defined as:

$$z(k) = \mathbb{I}\{\hat{y}^{(k)} = y^{(k)}\}$$

where $z(\mathrm{k}) = 1$(*Good*) indicates a correct prediction and $z(\mathrm{k}) = 0$(*Bad*) indicates an incorrect one. In practice $z(\mathrm{k})$ is constructed from inference files that contain, for each crop, at least the description, true label, predicted label, and membership split (train/validation/test). This logic is implemented directly in the monitor training scripts, where the label is created as an equality between predicted and true.

A crucial methodological aspect is that the monitor is trained on the same subject-independent splits already defined for the main model, thus maintaining the separation between train/validation/test even when moving from the multi-class problem to the binary Good/Bad problem. This avoids contamination between subjects and preserves the validity of the experimental protocol when measuring the (possible) effect of the monitor on the final aggregation.

5.4.2 Why a monitor can add information with respect to native confidence

The monitor is conceptually located at the same point in the pipeline: it does not modify the probability $\mathbf{p}(\mathrm{k})$ produced by the classifier, but attempts to estimate an additional quantity, interpretable as the probability of correctness [22]:

$$q(k) = \mathbb{P}(z(k) = 1 \mid \text{crop input})$$

The underlying hypothesis (and the motivation for the experiment) is that the error is not random: noise patterns, spectral ambiguities, or "borderline" configurations in the graph may leave a recognizable signature, and that a dedicated model can learn to distinguish regions of the input space where the main classifier tends to err, even when it is numerically "confident."

5.4.3 Monitor as a "linear probe" of the GNN backbone and variants

In terms of implementation, the main approach tested is a monitor based on the **same GNN architecture** used for clinical classification but adapted to two classes (Bad/Good). The implementation follows an extremely controlled transfer logic: the checkpoint of the main 3-class model is loaded, and a 2-class model is constructed in which all weights are copied except for the last classification layer, which is left free to adapt to the new task. Furthermore, to reduce capacity and the risk of overfitting (given the change of objective and the derived nature of the label), training is set up as linear probing: the entire network is frozen and only the last level is trained.

There is a specific reason for this choice: the backbone of the main model has already learned a useful representation of crops; instead of "re-learning the EEG," the monitor attempts to reuse that representation and estimate the probability of correctness as a simple separation in feature space. In this sense, the monitor is conceptually closer to a decision quality classifier than to a new diagnostic model.

Alongside this version, a variant with **ArcFace loss** [23] was also explored, where the last layer is replaced to produce an embedding and the Good/Bad separation is induced via angular margin, while maintaining the same idea: reuse of the backbone and training focused on a binary reliability objective.

Finally, in the meantime, we tried out an alternative approach with **CNN monitors** that work directly on the multichannel CWT representation, thinking that some errors might be more "visible" as time-frequency patterns (like residual artifacts or unstable spectral configurations) and that a 2D convolution could catch them effectively.

In these scripts, the Good/Bad target is built in the same way (match between predicted and actual), and the training also considers the possible imbalance between Bad and Good by weighting the loss.

In terms of code organization, the experiment is structured in a modular way:

- a "single experiment" script for training the GNN monitor and saving detailed predictions;
- a "multi-seed" version that automatically replicates the procedure on different seeds, reusing the inference files and the relative best-val checkpoint of the main model for each seed, and saving a CSV file with goodness (i.e., the probability of Good) for each split.

5.4.4 Critical review

It is important to clarify why this idea, although conceptually motivated, was kept as an exploratory experiment: the monitor is trained on a label derived from the behavior of the main model; therefore, it learns a form of "error map" that can be highly dependent on the checkpoint, seed, and generalization regime. For this reason, the multi-seed implementation was designed to verify the stability of the idea and to avoid any positive results being due to a lucky configuration. Furthermore, the experiment aims to address a sensitive issue that has already emerged in the probability analysis: the presence of "extreme" but incorrect predictions, which can have a significant impact at the subject level if not adequately controlled. In this interpretation, the monitor is not a simple accessory, but an attempt to introduce a reliability estimate complementary to native confidence, while keeping the overall structure of the pipeline unchanged.

6 Results

This chapter reports the quantitative results obtained with the pipeline described in Chapter 5, evaluating both **crop-level** performance and **subject/recording-level** performance, obtained by aggregating the predictions of crops belonging to the same subject. Two variants of the classification head are compared while keeping the GNN backbone unchanged:

- **Softmax head (multiclass)**: competitive classification with constraints $\sum_k p_k = 1$;
- **One-vs-Rest (OVR) head**: K independent binary heads with sigmoid, "winner-takes-all" decision on $\{p_k\}$.

All experiments are performed in 5 independent runs with seed (42, 1024, 1234, 2001, 2025), keeping the dataset, subject-independent split, and main training hyperparameters unchanged (Adam optimizer with $\mathrm{lr} = 10^{-5}$, weight decay 10^{-8}, scheduler disabled), always loading the best-val checkpoint for evaluation on the test.

6.1 Class distribution and assessment approach

The quantitative evaluation is conducted on three **subject-independent** splits (train, validation, test), shared by both architectures (Softmax and OVR). Table 1 shows the crop-level distribution of the **HC, FTD, and AD** classes in the three splits.

Table 1 - Distribution of classes in the three splits (crop-level).

Split	#HC	%HC	#FTD	%FTD	#AD	%AD	Total
Train	14 332	35,4%	9 158	22,6%	16 984	42,0%	40 474
Validation	4 781	32,6%	4 245	28,9%	5 640	38,5%	14 666
Test	4 985	34,0%	3 176	21,7%	6 493	44,3%	14 654

There is a moderate imbalance towards the **AD** class in all splits, with **FTD** systematically in the minority, especially in the test set. This has important implications:

- **accuracy** alone can be misleading (a classifier that favors AD can achieve decent accuracy but poor clinical utility on the FTD class);
- the most informative metric is **macro-F1** (unweighted average of the F1 scores of the three classes), which correctly penalizes poor performance on minority classes;
- it is also useful to discuss the **F1 scores by class**, because the difficulty is not uniform: typically, HC is more separable, while FTD falls in an "intermediate" zone between HC and AD.

6.2 Overall crop performance: Softmax vs OVR

Table 2 shows accuracy and macro-F1 (mean ± standard deviation over 5 seeds) for the two heads, on the three splits.

Table 2 - Accuracy and macro-F1 at crop-level (average ± std on 5 seeds).

Architecture	Split	Accuracy	Macro-F1
Softmax	Train	0,655 ± 0,031	0,623 ± 0,037
Softmax	Val	0,572 ± 0,005	0,530 ± 0,010
Softmax	Test	0,530 ± 0,015	0,463 ± 0,013
OVR	Train	0,636 ± 0,011	0,594 ± 0,018
OVR	Val	0,571 ± 0,011	0,522 ± 0,013
OVR	Test	0,544 ± 0,010	0,476 ± 0,008

Interpretation of results.

- **Training**. Softmax achieves higher values in training in terms of both accuracy and macro-F1 (0.655 vs. 0.636 and 0.623 vs. 0.594). This indicates a greater ability to adapt to training data, but also a potentially greater tendency toward overfitting.
- **Validation**. In validation, the two architectures are almost identical in accuracy (~0.57). Macro-F1 slightly favors Softmax (0.530 vs. 0.522), consistent with the fact that the original architecture was set with softmax head as default.
- **Test**. On the test set, the picture is reversed: **OVR** achieves higher accuracy (0.544 vs. 0.530) and better macro-F1 (0.476 vs. 0.463), with lower standard deviation. In other words, OVR shows more **robust generalization** on seeds.

A useful indicator is the **train-test gap** in macro-F1:

- Softmax: $0{,}623 \rightarrow 0{,}463 (\Delta \approx 0{,}16)$

- OVR: $0{,}594 \rightarrow 0{,}476 (\Delta \approx 0{,}12)$

Softmax “explains” the train better but loses more performance on unseen data; OVR is more conservative on the train and degrades less on the test. This behavior is consistent with the decision geometry: Softmax imposes competition on the simplex, while OVR learns more decoupled “class vs. rest” binary margins on the same global embedding.

6.2.1 Class analysis at crop level

To understand where OVR's advantage comes from, it is essential to analyze the **F1 by class**. Table 3 shows the average F1s (out of 5 seeds) for HC, FTD, and AD in validation and testing.

Table 3 - F1 by class (average of 5 seeds, crop-level).

Architecture	Split	F1(HC)	F1(FTD)	F1(AD)
Softmax	Val	0,655	0,306	0,629
Softmax	Test	0,610	0,209	0,568
OVR	Val	0,651	0,278	0,637
OVR	Test	0,606	0,231	0,592

HC class. The two architectures are practically indistinguishable: F1 ~0.65 in validation and ~0.61 in testing. This suggests that healthy control is relatively “easy” for the backbone in both cases.

FTD class (minority and more difficult). In validation, Softmax is slightly better (0.306 vs. 0.278), but in testing, the situation is reversed: OVR achieves F1 ≈ 0.231 compared to 0.209 for Softmax. Looking also at **precision** and **recall** on the test for FTD:

- Softmax (FTD): precision ≈ 0.35, recall ≈ 0.15
- OVR (FTD): precision ≈ 0.41, recall ≈ 0.16

OVR reduces FTD false positives (higher precision) and slightly improves recall, increasing F1 on a class that is both minority and clinically sensitive.

AD Class. In validation, OVR is already slightly ahead (0.637 vs. 0.629), and in the test, the advantage is more evident (0.592 vs. 0.568). In terms of precision/recall on the test:

- Softmax (AD): precision ≈ 0.58, recall ≈ 0.56
- OVR (AD): precision ≈ 0.58, recall ≈ 0.61

With equal precision, OVR recovers more AD cases, reducing false negatives.

Crop-level summary. With the same backbone, OVR seems to manage pathological classes (**FTD** and **AD**) better in the subject-independent test, while HC remains stable. This is precisely the desirable behavior in a clinical setting, where performance on pathological classes is the most critical aspect.

6.2.2 Analysis of probabilities and decision margins (crop-level)

Thanks to inference files, it is also possible to analyze predicted probability distributions. For each crop c, the following are defined:

- maximum probability assigned by the model:

$$p_{\text{top}}(k) = \max_{t} \ p_t(k)$$

- margin between first and second class most likely:

$$m(k) = p_{\mathrm{top1}}(k) - p_{\mathrm{top2}}(k)$$

By aggregating on the test set and the five seeds, we observe that:

Softmax head:

- correct predictions:
  - $\mathbb{E}[p_{\mathrm{top}}] \approx 0{,}63$
  - $\mathbb{E}[m] \approx 0{,}39$
- wrong predictions:
  - $\mathbb{E}[p_{\mathrm{top}}] \approx 0{,}56$
  - $\mathbb{E}[m] \approx 0{,}28$
- approximately **8%** of crop have $p_{\mathrm{top}} \geq 0{,}9$; among these, ~**15%** are still incorrect.

OVR head

- correct predictions:
  - $\mathbb{E}[p_{\mathrm{top}}] \approx 0{,}68$
  - $\mathbb{E}[m] \approx 0{,}48$
- wrong predictions:
  - $\mathbb{E}[p_{\mathrm{top}}] \approx 0{,}60$
  - $\mathbb{E}[m] \approx 0{,}35$
- approximately **15%** of crop ha $p_{\mathrm{top}} \geq 0{,}9$; among these, ~**22%** are incorrect.

Two complementary messages can be drawn from these figures.

1. **On average, decisions are more “clear-cut” with OVR**. OVR produces higher average probabilities and margins, both for correct and incorrect decisions. This is consistent with the nature of the loss (BCE on margins per class) and with the absence of the global normalization constraint of Softmax.

2. **Greater overconfidence in extreme predictions.** OVR increases the proportion of predictions with $p_{\text{top}} \geq 0{,}9$, but the percentage of errors among these also increases. In other words, OVR is more "decisive" even when it is wrong. This feature becomes crucial when moving from the crop unit to the subject unit: if the aggregation is well designed, it can transform locally noisy decisions into a robust global diagnosis; conversely, if aggregation and calibration are inadequate, overconfidence can become a risk.

6.3 Subject aggregation: weighted soft voting

Clinically, the decision unit is the subject (or the individual record). For this reason, an aggregation strategy that exploits probabilistic information is implemented: weighted soft voting.

Let's consider a dataframe of inferences with:

- group_col: subject/recording identifier;
- softmax_col: probability vector for crop (Softmax for the Softmax head; sigmoid OVR possibly reported to a vector that can be used for voting);
- true_col: truth label at crop level.

For each crop k, the weight is defined as:

$$w_k = \max_t \, p_t(k)$$

that is the confidence associated with the winning crop class. For a subject i with crop set $\mathcal{K}_i$, the weighted sum of probabilities is calculated:

$$\mathbf{p}_i = \sum_{k \in \mathcal{K}_i} w_k \; \mathbf{p}(k)$$

where $\mathbf{p}(k)$ is the probability vector of the crop. The prediction at the subject level is:

$$\hat{y}_i = \arg\max_t \mathbf{p}_i[t]$$

Equivalently, weight can be interpreted as:

$$w_k = \hat{P}_{k,\hat{y}^{(k)}}$$

where $\hat{y}^{(k)}$ is the predicted class for the crop and $\hat{P}_{k,\hat{y}^{(k)}}$ is its confidence. Therefore, "certain" crops contribute more to the score, while uncertain crops (almost uniform probabilities) have less impact.

Extension of weighted soft voting: hybrid fusion with monitor confidence

In the scheme described confidence w_k reflects the "strength" of the main model's prediction, but not necessarily its actual reliability: numerically very confident predictions may still be incorrect, especially in the presence of ambiguous crops or crops that are out-of-distribution with respect to the learned patterns.

To mitigate this limitation, a variant was explored in which the weight assigned to each crop does not depend solely on the native confidence of the classifier but also integrates an explicit reliability estimate produced by an auxiliary model, called a **monitor**. The monitor tackles a binary Good/Bad task and produces, for each crop, a quantity $q(k)$ that can be interpreted as the probability that the main model's prediction is correct:

$$q(k) \approx \mathbb{P}\left(\hat{y}^{(k)} = y^{(k)} \mid k \right)$$

The goal is not to replace the classifier, but rather to use $q(k)$ as a complementary signal to modulate the impact of individual crops in the subject-level aggregation.

In particular, a **hybrid weight** is defined as a combination of the confidence of the main model and the confidence of the monitor:

$$\widetilde{w}_k(\alpha) = \alpha\, w_k + (1-\alpha)\, q(k), \alpha \in [0,1]$$

By replacing w_k with $\widetilde{w}_k(\alpha)$, the aggregation rule retains the same structure, but with a weight that also incorporates the accuracy estimate:

$$\mathbf{p}_i = \sum_{k \in \mathcal{K}_i} \widetilde{w}_k\,(\alpha)\, \mathbf{p}(k), \hat{y}_i = \arg\,\max_t\, \mathbf{p}_i[t]$$

In this formulation, w_k continues to represent the intensity with which the main model supports the predicted class via $\mathbf{p}(k)$, while $q(k)$ acts as a reliability factor: crops that the monitor considers likely to be correct contribute more to the weighted sum, while suspicious crops are attenuated even if locally confident.

The parameter α regulates the balance between the two signals. For $\alpha \to 1$ the method coincides with standard weighted soft voting; for $\alpha \to 0$ the modulation of the crop contribution is guided almost entirely by the monitor. Consistent with the experimental protocol adopted in the thesis, the selection of α is conceived as a tuning procedure on the **validation set**, choosing the value that maximizes the subject-level metrics and reserving the test set for the final evaluation only to avoid optimization bias.

6.4 Subject-level results on test set

The test set contains **19 subjects**, so the possible accuracies are multiples of:

$$\frac{1}{19} \approx 5{,}26\%$$

Table 4 shows the results of weighted soft voting on the test, averaging over the five seeds, for four configurations:

- **Softmax (5_seed):** “optimized” subject-independent split (the main one) with Softmax head;
- **Softmax (5_seed_CV):** cross-validation with random assignment of subjects to training/validation based on seed;
- **OVR (5_seed_ovr):** main split with OVR head;
- **OVR (5_seed_ovr_CV):** cross-validation with OVR head.

Table 4 — Weighted soft voting on the test set (19 subjects, mean ± std on 5 seeds).

Setup	Accuracy [%]	F1-score [%]	Precision [%]	Recall [%]
Softmax (5_seed)	57.9 ± 5.8	49.1 ± 4.9	45.2 ± 2.8	57.9 ± 5.8
Softmax (5_seed_CV)	55.8 ± 5.4	48.3 ± 6.4	47.8 ± 12.8	55.8 ± 5.4
OVR (5_seed_ovr)	66.3 ± 5.4	63.3 ± 5.4	71.9 ± 5.6	66.3 ± 5.4
OVR (5_seed_ovr_CV)	48.4 ± 5.2	40.4 ± 4.8	36.5 ± 3.9	48.4 ± 5.2

Key observations.

In the main configuration (rows “5_seed” and “5_seed_ovr”), the difference is clear:

- OVR gains **+8.4** points in subject-level accuracy compared to Softmax (66.3% vs. 57.9%);
- The subject-level macro-F1 score increases from ~49.1% (Softmax) to ~63.3% (OVR), an increase of approximately **+14** points;
- The average precision increases from ~45.2% to ~71.9% (≈ **+27** points), suggesting that the subject-level diagnoses produced by OVR are much less affected by false positives than Softmax.

Even at the individual seed level, OVR's superiority is frequent: for example, for **seed 1234**, OVR achieves **73.68%** accuracy and **68.59%** F1, compared to **57.89%** and **50%** with Softmax on the same split.

Observations on CV configurations.

CV configurations act as a control: subjects are randomly assigned to train/validation, and with a limited number of subjects, performance becomes more sensitive to the particular partition. In this scenario:

- Softmax maintains values similar to the main configuration;
- OVR shows a marked reduction (average accuracy ~48.4%).

This suggests that:

- the advantage of OVR is more evident when the partitioning of subjects is **reasoned/optimized** (main setting, more clinically realistic);
- OVR may be more sensitive to random imbalances in the composition of train/validation, because each binary "class vs. rest" head requires adequate coverage of the class in training.

6.5 Why OVR benefits more from voting than Softmax

The subject-level results show that OVR benefits much more from weighted soft voting than from Softmax. This behavior can be interpreted in three ways:

- **Quality of probabilities per class at crop level.**

  At window level, OVR:

  - improves the F1 of FTD and AD in tests;
  - increases the sensitivity of AD at the same precision;
  - produces wider average margins when the prediction is correct.

In an AD or FTD subject, this translates into many crops that vote correctly with high confidence: in the weighted sum, these crops accumulate consistent evidence for the correct class, overcoming the noise of ambiguous or occasionally incorrect crops.

- **Weight structure in weighted soft voting.**

  The aggregation rule gives more weight to the most confident crops.

  For Softmax, the probabilities are on average more "flat" and the separation between correct and incorrect in terms of confidence is less clear-cut; as a result, the sum $\mathbf{p}_i$ tends to be less polarized and the evidence of the true class can be "diluted" by many moderately confident votes for incorrect classes.

  For OVR, on the other hand, in pathological classes, the corresponding head often produces higher p_k margins: correct crops therefore have greater w_k weights and $\mathbf{p}(k)$ vectors that are more biased towards the correct class. This creates a very effective accumulation mechanism.

- **Decoupling margins and specialization by class.**

  In softmax, the constraint $\sum_k p_k = 1$ forces competition: increasing the probability of one class implies reducing the others. This can act as regularization, but in the presence of phenomenologically overlapping classes (HC-FTD-AD), it can lead to compromises (e.g., FTD "squashed" between HC and AD).

  In the OVR model, each head r optimize the "r vs rest." margin independently: at the subject level, the weighted sum integrates evidence that is already "per-class" and partially decoupled. If a subject is AD, the AD head can produce a consistent sequence of crops with high p_{AD}, while the other heads remain more neutral; the subject-level vote therefore behaves as a temporal integral of the evidence per class.

Two levels of integration in the pipeline.

The combination of “graph-centric GNN + OVR head + weighted soft voting” achieves two types of integration:

1. **spatial and spectral** through GNN (electrode topology + CWT pattern),
2. **long-range temporal** through weighted summation over subject windows.

Softmax also benefits from aggregation (subject-level performance improves compared to crop-level), but OVR seems to make the most of the information distributed across many crops, amplifying the difference between consistent pathological evidence and local noise.

6.6 Summary of the chapter

The results show that:

- at the crop level, Softmax and OVR are similar in validation, but OVR **generalizes better** in testing, especially on pathological classes (FTD and AD);
- at the subject level, weighted soft voting **amplifies** the advantage of OVR, bringing the average accuracy to 66.3% and macro-F1 to 63.3% in the main split;
- the CV configurations indicate that OVR is more sensitive to random subject partitions, likely due to the “per-class” nature of binary heads and the limited size of the subject-level sample.

7 Discussion and conclusions

This chapter interprets the experimental results in light of the methodological choices introduced in the previous chapters, discussing their significance in the clinical and computational context. The main limitations of the study and possible future developments are also highlighted.

7.1 What really comes out

The objective of this thesis was to evaluate whether a **graph-centric** pipeline based on **time-frequency representations (CWT)** and **Graph Neural Networks** is capable of discriminating between **HC**, **FTD**, and **AD** using the clinical dataset of Miltiadous et al. The evaluation considered two levels, both of which are relevant:

- **Crop-level (1 s):** measures the discriminatory capacity on individual windows, useful for understanding whether the model "sees" local markers and for analyzing the nature of confusion between classes;
- **Subject-level:** this is the clinical decision unit, and therefore the most significant metric from an application perspective.

The first solid result is that, at crop level, Softmax and OVR are essentially comparable in validation (accuracy ~0.57), but **OVR generalizes better in test** in terms of accuracy and macro-F1. This behavior suggests that the Softmax head, despite performing better on training, tends to "close" more on the configurations observed in training, while OVR assumes a more conservative behavior and is less subject to degradation on unseen data. This difference is also evident in the **train-test gap**, which is smaller for OVR.

At the per-class level, the stability of the **HC** class is an important aspect: both heads reach similar and relatively high values, indicating that the separation between "healthy vs. pathological" is simpler than the discrimination between the two pathological conditions. However, it is in the distinction between **FTD** and **AD** that the most interesting effect is observed: OVR recovers performance especially on the pathological classes in the test, in particular improving F1 on FTD and increasing recall on AD without losing precision. This is consistent with the idea that FTD and AD share portions of phenomenological space (partially overlapping EEG patterns), and that a model with more ‘specialized’ boundaries per class can better handle the overlap.

A second key finding is that subject-level aggregation via **weighted soft voting** significantly amplifies the benefits of OVR. This observation is not just a “numerical effect,” but reflects a structural property of the pipeline: the model produces predictions on very short windows (1 s), so it is natural for individual windows to be noisy or ambiguous; However, on a recording composed of many crops, the diagnostic signal tends to be **redundant** and **repeated** over time. Aggregation therefore allows the diagnosis to be estimated as a form of **temporal integration of evidence**, conceptually closer to clinical reasoning (which is not based on a single second of EEG).

Weighted voting, specifically, mitigates the influence of “uncertain” crops and enhances those with greater confidence. Since OVR produces higher average margins and probabilities (albeit with a greater risk of local overconfidence), this structure translates into a more effective accumulation of correct evidence at the subject level, especially for pathological classes. The empirical result is consistent: OVR achieves ~66% subject-level accuracy in its main configuration with macro-F1 > 63%, with a notable increase in precision compared to Softmax.

7.2 Why the combination of CWT + graph + GNN makes sense

The thesis assumes that diagnostic information is not only “how much” power there is in a band, but also includes:

- **time-frequency structures** (peaks, bursts, transitions, non-stationary variations);

- **spatial patterns** (distribution across regions and asymmetries);
- **relational patterns** (interactions and co-variations between channels).

CWT preserves information that is richer than simple average power in the band, and GNN allows the EEG to be naturally modeled as a system of connected sensors. Although in this work the arcs are not estimated from subject-specific functional connectivity, the sparse topology imposes a structure that allows message passing to synthesize "local" information (per electrode) into a global representation of the crop, while keeping complexity under control.

The choice of static learnable arc weights (independent of features) lies somewhere between two extremes: on the one hand, completely fixed graphs, and on the other, complex attention mechanisms that are potentially more unstable on small datasets. The learnable α weights allow the model to globally "reweight" the connections, introducing an interpretative element and structural adaptation without exploding the number of parameters.

7.3 Why subject-independent splitting is crucial

A crucial point, often overlooked in EEG-based studies, is that window segmentation creates many samples derived from the same subject. If the partition between training and testing is not subject-independent, the model may achieve artificially high performance by learning idiosyncratic patterns of the subject (identity leakage), rather than generalizable biomarkers.

In this thesis, the split is explicitly subject-independent, which makes the results more realistic. The presence of CV configurations with random subject assignment also shows that, with small datasets, the specific partition can significantly affect performance. In particular, the sensitivity of OVR in CV (lower and more variable performance) reinforces the idea that, in multi-class scenarios with minority classes, class coverage in training and set composition are decisive.

7.4 Limitations

Despite its results and methodological consistency, the study has some significant limitations, which must be clarified in order to interpret the evidence correctly.

- **Small dataset size and limited number of subjects in the test.**
  The test set contains 19 subjects; this implies that subject-level accuracy can only vary in multiples of ~5.26%. Therefore, differences of a few percentage points can be influenced by a very small number of errors in either direction. The robustness estimate should ideally be supported by a larger number of subjects or multi-site validation.

- **Class imbalance and intrinsic difficulty of FTD.**
  The FTD class is a minority class and phenomenologically “intermediate.” Even when the model improves F1 on FTD, the estimate remains fragile: small changes in the decision threshold or subject partition can significantly shift precision/recall. This is also visible in cross-validation behavior.

- **Sensitivity to partitions (instability in CV).**
  The OVR configuration shows an advantage in the “optimized” setting but worsens markedly with random partitions. This indicates that the model may be more sensitive to the composition of the training set, especially since binary heads per class require sufficient and representative examples of each category.

- **Overconfidence of the OVR head at crop level.**
  Probability analysis shows that OVR produces more “peaked” predictions, but with a significant number of errors even for $p_{top}\geq0.9$. In the absence of explicit calibration (e.g., temperature scaling or isotonic regression on validation), probabilities should not be interpreted as calibrated estimates of clinical risk. Subject-level voting mitigates this problem but does not eliminate it.

- **A priori defined graph and absence of subject-specific functional connectivity.**
  The topology is neuro-plausible but does not derive from a direct estimate of connectivity (coherence/PLV, etc.). This reduces complexity and increases stability but may limit the model's ability to capture subject- or condition-specific network biomarkers. A natural extension would be to integrate band-width connectivity-weighted graphs or multi-graphs.

- **Cross-device/cross-site generalizability not verified.**
  Recordings come from a single clinical setting and instrumentation. The robustness of the model to variations in mounting, impedances, acquisition protocols, or different populations has not been tested. This is crucial for any translational perspective.

7.5 Future developments

Based on these limitations, concrete areas of work emerge:

- **Probability calibration**, especially for OVR, using post-hoc validation techniques (temperature scaling, Platt scaling per head, isotonic regression), with Expected Calibration Error/Brier score analysis.
- **Functional connectivity and adaptive graphs**, integrating arc weights derived from coherence/PLV in specific bands or multi-layer graphs, possibly combined with anatomical topology.

- **Ablation study**, isolating the contribution of CWT vs. alternatives (PSD, STFT); temporal compression; type of pooling; static edge weights vs. attention.
- **External validation** on independent datasets (cross-site) or through domain adaptation strategies, to estimate robustness to domain shifts.

7.6 Conclusions

This thesis proposed and evaluated a pipeline for the automatic diagnosis of dementia (HC/FTD/AD) from resting-state EEG, based on CWT representations and Graph Neural Networks. The results show that:

- the graph-centric GNN is able to learn discriminative patterns in the time-frequency domain and integrate spatial/relational information between electrodes;
- the OVR head, while producing locally more "decisive" predictions, shows better generalization in the crop-level test and a marked advantage when predictions are integrated at the subject level via weighted soft voting;
- subject-level aggregation is essential for transforming predictions on short windows into a clinically meaningful decision and constitutes a second level of integration complementary to message passing on the graph.

Overall, the work highlights that, in a realistic subject-independent setting and on a public clinical dataset that also includes FTD, the combination of **CWT + GNN + OVR + weighted voting** represents a promising approach consistent with the interpretation of AD and FTD as network disorders. At the same time, the limitations discussed indicate that the robustness of the method needs to be further validated on larger, multi-site samples, with particular attention to probability calibration and stability with respect to subject partitions.

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